

Goal: If a quantity changes over time, explain how its derivative changes over time.

1 Introduction (Circles)

Problem 1. *A circle's area and radius are related by the equation:*

$$A = \pi r^2$$

If A and r are changing as time advances, is there any relation between $\frac{dA}{dt}$ and $\frac{dr}{dt}$?

Problem 2. *(a) In the previous example, if $r = 2$ and $\frac{dr}{dt} = 3$, then what is the value of $\frac{dA}{dt}$? Interpret.*

Lecture 14: Related Rates (Perimeter/Area/Volume Relations)

(b) In the previous example, if $r = 1$ and $\frac{dA}{dt} = 2\pi$, then what is the value of $\frac{dr}{dt}$?

Problem 3. The radius of a circle r is increasing at a constant rate 3 cm/min.

(a) Find the rate of change of the area of the circle (A) when the radius is 5cm.

- (b) Find the rate of change of the circumference of the circle (C) when the radius is 5cm.

2 Rectangular Prisms

Problem 4. *The edges of a cube are shrinking at a rate of 10 cm/s.*

- (a) How fast is the volume (V) shrinking when each side length is 9cm long?

- (b) How fast is the surface area (A) shrinking when each side length is 9cm long?

3 Spheres

Problem 5. *A balloon is (roughly) a sphere. The balloon deflates and its radius decreases at a rate of 2 cm/s.*

- (a) How fast is the volume (V) shrinking when the radius is 5cm long?

(b) How fast is the surface area (A) shrinking when the radius is 5cm long?

4 Cylinders

Problem 6. *A cylindrical tank with a radius and height of 100 cm stands upright. Water is being drained at a rate of $7\text{cm}^3/\text{s}$. How fast is the water level changing when the tank is half empty.*

5 Cones

Problem 7. *Sand pours onto a surface at $15\text{cm}^3/\text{s}$, forming a conical pile with a base diameter that is always equal to the pile's altitude. How fast is the altitude of the pile increasing when the pile is 8cm high?*